Course 2: Enterprise ASP.NET MVC 5 & SQL Server

ASP.NET MVC 5 Enterprise Architecture & Web API Reference

Official Study Guide • Instructor: Dr. Vikram Sen (Principal Cloud Solutions Architect) • EduPulse Academy

1. HTTP Request Lifecycle & OWASP Security Defenses

Every incoming HTTP request in ASP.NET MVC traverses HttpApplication, RoutingEngine, ControllerFactory, ActionFilters, and ViewEngine.

Cross-Site Request Forgery (CSRF) must be guarded against on all state-altering POST/PUT requests using [ValidateAntiForgeryToken] alongside @Html.AntiForgeryToken().

Idempotency guarantees that issuing identical HTTP requests (GET, PUT, DELETE) produces identical server states, whereas POST submissions create fresh records and must be shielded with PRG (Post-Redirect-Get) patterns.

```
[HttpPost]
[ValidateAntiForgeryToken]
public ActionResult SubmitAssignment(StudentSubmission model)
{
    if (!ModelState.IsValid) return View(model);
    _repo.Save(model);
    return RedirectToAction("Success"); // PRG Pattern
}
```

2. Data Access Architecture: Repository & ADO.NET Parameters

Direct string concatenation into SQL queries exposes applications to critical SQL Injection vulnerabilities. Always employ parameterized SqlCommand parameters with explicit SqlDbType declarations.

Transaction scopes ensure atomic ACID execution across multi-table insertions such as Order Header and Order Line items.

```
using (SqlCommand cmd = new SqlCommand("SELECT * FROM LMS.Users WHERE Email = @Email", conn))
{
    cmd.Parameters.Add("@Email", SqlDbType.NVarChar, 100).Value = email;
    // Executes safely with parameter tokenization
}
```

3. Unit Testing & Dependency Inversion Principle (SOLID)

Controllers must never directly instantiate ADO.NET connections or concrete repositories. Inject repository interfaces via constructor injection to allow mock implementations during NUnit/xUnit testing.

Action filter attributes encapsulate cross-cutting concerns including authentication, role gating, and structured exception logging.

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